THE GIANT STEPS

- New weapon systems have to be continuously developed for conflicts in a state of flux
- Advanced assault rifles, howitzers, better cybersecurity, strategic mobility are things needed by the Indian armed forces
- Strengthening defences against cyber threats and developing offensive cyber capabilities are equally important
- Technology by itself won't win wars. The human element in war remains vital

learning from experiences of the present and extrapolating these for the future.

The most important facet is to be able to identify and then occupy or control the proverbial 'high ground', so essential for victory. In the past, the side that dominated the hill tops was decidedly at an advantage, as it not only made their positions more secure, it also enabled them to see what lay on the 'other side of the hill'. This 'high ground' moved on to domination of the air space above the battlefield, no longer remaining a terrestrial construct. Domination of the air became a prerequisite for the conduct of successful ground operations. Where do we further move on to?

The arena in the future may well be space, including cyberspace. Efforts to develop space-based weapon systems and Anti-Satellite Systems (ASAT) are well under way. So are efforts to develop Near Earth Orbit and High-Altitude Pseudo Satellites (HAPS). All these are pointers to where the battle might be fought in the future. In this battle, cyber warfare will have a major role to play. The basic strategy has not changed, only the place and manner in which it is to be applied has. India has created a tri-service cyber agency headed by a two-star general or equivalent. Perhaps this would be the right time to upgrade it to a command as part of the theaterisation process.

AI-enabled unmanned systems: The impact of technological developments, including quantum technologies and artificial intelligence (AI), on our war fighting methodology and the weapon platforms in our inventory necessitates a review. The dynamics of the battlefield are in a never-ending state of flux. Newer weapon systems have to be developed and existing platforms have to adapt to changing conditions. While changes in tactics, techniques and procedures may mitigate deficiencies in current platforms, a hard look is required on the future of main battle tanks, aircraft carriers and other large surface combatants, and manned aircraft, drawing from the lessons of current conflicts. The future will lie in larger numbers of low-cost AI-enabled unmanned systems. Current platforms with contemporary add-ons and countermeasures have become too unwieldy. Moreover, the high cost of current platforms will make commanders hesitant to commit them to battle. The Indian Army's modernisation efforts should focus on key areas, aimed at making it more agile, technologically advanced and capable of addressing diverse security challenges. For technology/ equipment upgrades, the army needs to replace outdated rifles with modern assault rifles/ carbines. It also has to acquire advanced howitzers and rocket systems and enhance anti-aircraft and anti-drone capabilities. An upgrade to secure, integrated communication networks is crucial too.

Cybersecurity has gained increased salience in recent decades. Thus, strengthening defences against cyber threats and developing offensive cyber capabilities are equally important. Strategic mobility and logistics determine a military's defensive and offensive capabilities, and improving rapid deployment and sustainment capabilities and enhancing infrastructure along border areas are important.

The armed forces operate with immensely skilled professionals. Training and skill development for soldiers, especially for Agniveers, and leadership development and joint operations training for others, are vitally important.

Investing in indigenous defence technologies reduces dependence on foreign arms, and collaborating with research institutions and the private sector can achieve that. For achieving jointness and integration, there should be coordination of operations with other armed forces, enhancement of interoperability, and theaterisation.

Technology will no doubt shape the character of war. However, there is a need to balance human resources and technology. In spite of technological developments over the centuries, 'boots on ground' remain critical, as the primacy of land will remain paramount.

In the words of historian T.R. Fehrenbach, "You may fly over a land forever; you may bomb it, atomise it, and wipe it clean of life, but if you desire to defend it...you must do this on the ground, the way the Roman legions did—by putting your soldiers in the mud." This basic truth has not changed. The ongoing war in Ukraine is a testament to this fact. Our wars will be fought in the sand, mud and snow; in deserts, plains, mountains and jungles; over the land, sea and air, by our soldiers, sailors and airmen, with victory measured in terms of square kilometres won or lost.

It is human to focus on the known. The need of the hour is for military strategists and planners to look for answers in the proverbial blind spot, the unknown, which will only be possible by shedding entrenched concepts and overcoming prejudices. Over-reliance on any one facet, particularly the feeling that technology by itself will win wars, is misplaced. Notwithstanding our imperfections, the day technology supplants humans will be the end of humanity as we know it. ■

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